

CI/CD Workflow Report

GitHub and Jenkins Pipeline Implementation

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Roll No.	2301730325
Experiment	DevOps Git Lab
Tools	Git, GitHub, Jenkins
Pipeline	devops-git-lab-pipeline
Repository	Amit-M-14/devops-git-lab

This report documents the CI/CD workflow implemented using GitHub as the source-code repository and Jenkins as the automation server.

1. Objective

The objective of this experiment is to implement a basic CI/CD workflow in which source code is maintained in GitHub and Jenkins retrieves the repository, reads the Jenkinsfile, and executes automated Build, Test, and Deploy stages.

2. CI/CD Pipeline Architecture

1. Developer / Git	Source changes are committed to the repository.
2. GitHub	Stores the project files and Jenkinsfile.
3. Jenkins SCM	Jenkins connects to GitHub and checks out the selected branch.
4. Build	The pipeline executes the build stage.
5. Test	The pipeline executes the test stage.
6. Deploy	The pipeline executes the deployment stage.
7. Result	Jenkins reports SUCCESS when all stages complete.

Workflow: Git/GitHub → Jenkins SCM → Checkout → Build → Test → Deploy → SUCCESS

3. Jenkins Pipeline Configuration

Jenkins Job	devops-git-lab-pipeline
SCM	Git
Repository	https://github.com/Amit-M-14/devops-git-lab
Branch	master
Script Path	Jenkinsfile
Pipeline Type	Pipeline script from SCM

The Jenkins job was configured to obtain its pipeline definition from the GitHub repository. The Jenkinsfile is stored at the repository root, and the master branch is used for the experiment.

4. Jenkinsfile and Pipeline Stages

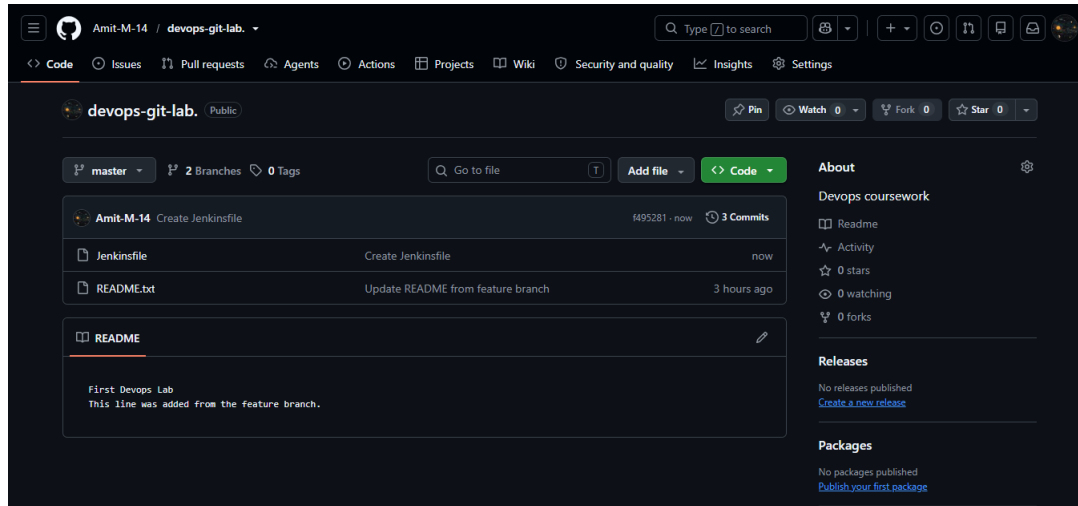
```
pipeline {
    agent any
    stages {
        stage('Build') {
            steps {
                echo 'Building the project...'
            }
        }
        stage('Test') {
            steps {
                echo 'Testing the project...'
            }
        }
    }
}
```

```
}
stage('Deploy') {
    steps {
        echo 'Deploying the project...'
    }
}
post {
    success {
        echo 'CI/CD Pipeline completed successfully!'
    }
}
}
```

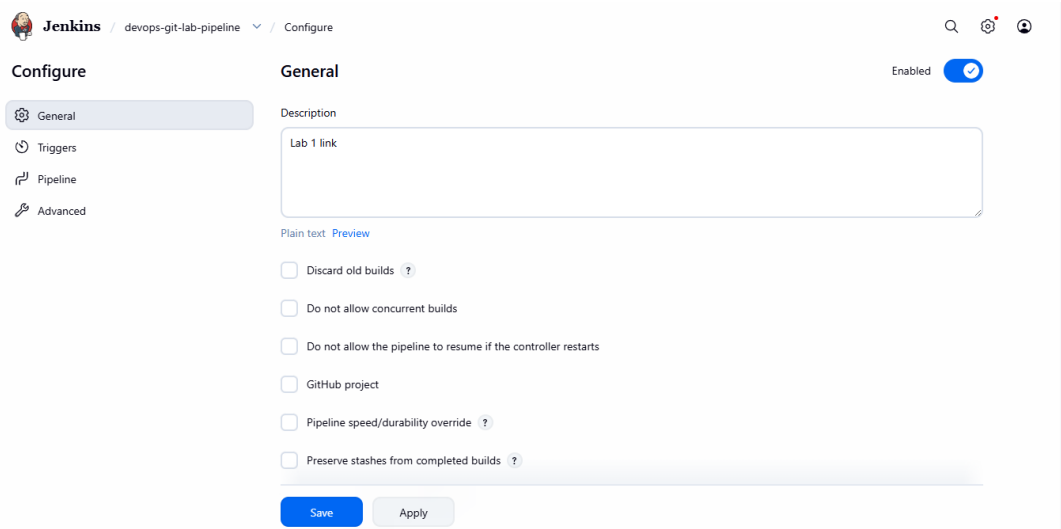
5. Pipeline Execution and Automation Results

Jenkins successfully retrieved the Jenkinsfile from GitHub, initialized the workspace, fetched the repository, checked out the master branch, and executed the pipeline stages.

GitHub repository containing the Jenkinsfile



Jenkins SCM configuration



Repository URL and credentials configuration

Repository URL ?

https://github.com/Amit-M-14/devops-git-lab.

Credentials ?

- none -

+ Add

Advanced ▼

Branch and Jenkinsfile script path configuration

Branch Specifier (blank for 'any') ?

*/master

+ Add Branch

Repository browser ?

(Auto)

Additional Behaviours

+ Add

Script Path ?

Jenkinsfile

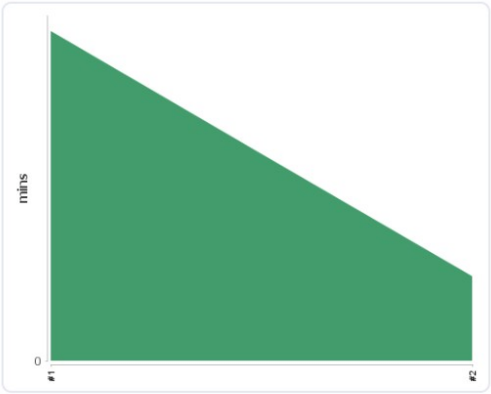
☒ Lightweight checkout ?

Successful build history

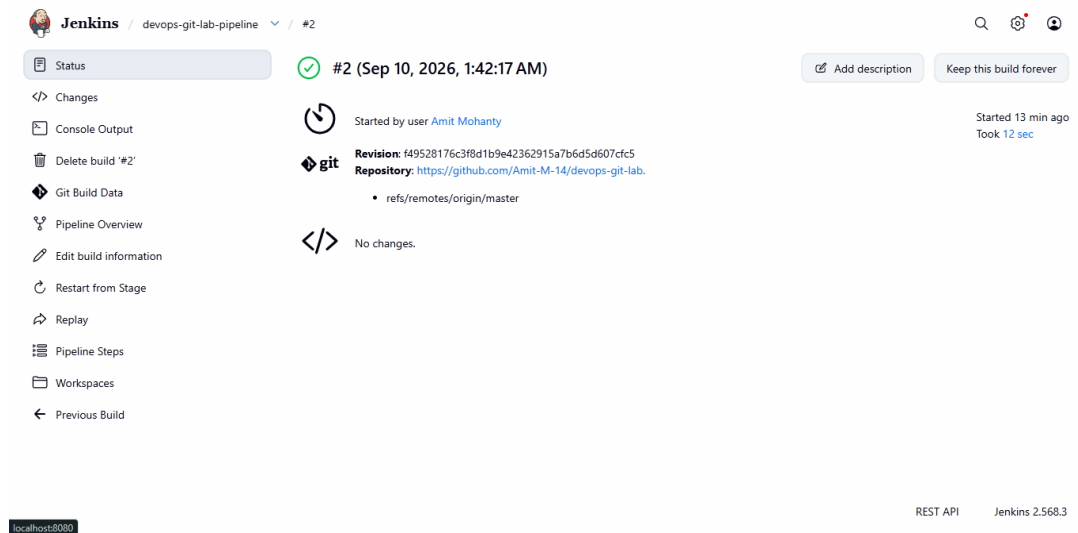
Build Time Trend

S	Build ↑	Time Since	Duration	
✓	#2	27 min	12 sec	📄
✓	#1	32 min	48 sec	📄

Icon: S M L



Jenkins Build #2 status and Git revision



The screenshot shows the Jenkins web interface for build #2 of the 'devops-git-lab-pipeline'. The build is in a 'Success' state, indicated by a green checkmark. The build was started by user 'Amit Mohanty' on September 10, 2026, at 1:42:17 AM. The console output shows the build process, including the checkout of the Jenkinsfile from a GitHub repository. The build took 12 seconds and started 13 minutes ago. The interface includes a sidebar with navigation links for Status, Changes, Console Output, Delete build, Git Build Data, Pipeline Overview, Edit build information, Restart from Stage, Replay, Pipeline Steps, Workspaces, and Previous Build. The bottom status bar shows 'localhost:8080', 'REST API', and 'Jenkins 2.568.3'.

Jenkins / devops-git-lab-pipeline / #2

Status ✓ #2 (Sep 10, 2026, 1:42:17 AM) Add description Keep this build forever

Changes

Console Output

Delete build '#2'

Git Build Data

Pipeline Overview

Edit build information

Restart from Stage

Replay

Pipeline Steps

Workspaces

Previous Build

Started by user [Amit Mohanty](#)

Revision: [f49528176c3f8d1b9e42362915a7b6d5d607cfc5](https://github.com/Amit-M-14/devops-git-lab)

Repository: <https://github.com/Amit-M-14/devops-git-lab>

- refs/remotes/origin/master

No changes.

Started 13 min ago
Took 12 sec

localhost:8080 REST API Jenkins 2.568.3

Console output showing GitHub checkout and Jenkinsfile retrieval

```
Started by user Amit Mohanty
Obtained Jenkinsfile from git https://github.com/Amit-M-14/devops-git-lab.
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\devops-git-lab-pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
> git.exe rev-parse --resolve-git-dir C:\ProgramData\Jenkins\.jenkins\workspace\devops-git-lab-pipeline\.git #
timeout=10
Fetching changes from the remote Git repository
> git.exe config remote.origin.url https://github.com/Amit-M-14/devops-git-lab. # timeout=10
Fetching upstream changes from https://github.com/Amit-M-14/devops-git-lab.
> git.exe --version # timeout=10
> git --version # 'git version 2.53.0.windows.2'
> git.exe fetch --tags --force --progress -- https://github.com/Amit-M-14/devops-git-lab.
+refs/heads/*:refs/remotes/origin/* # timeout=10
> git.exe rev-parse "refs/remotes/origin/master^{commit}" # timeout=10
Checking out Revision f49528176c3f8d1b9e42362915a7b6d5d607cfc5 (refs/remotes/origin/master)
> git.exe config core.sparsecheckout # timeout=10
> git.exe checkout -f f49528176c3f8d1b9e42362915a7b6d5d607cfc5 # timeout=10
Commit message: "Create Jenkinsfile"
```

Console output showing Build, Test, Deploy and successful completion

```
Testing the project...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploying the project...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
CI/CD Pipeline completed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Successful Jenkins build summary

Finished: SUCCESS — confirmed in the console output above, following the completion of the Build, Test and Deploy stages.

6. Observations

- The GitHub repository contains the Jenkinsfile required for the pipeline.
- Jenkins was configured with Git SCM and the repository URL.
- Jenkins successfully obtained the Jenkinsfile from GitHub and checked out the master branch.
- The Build, Test, and Deploy stages executed successfully.
- The final console output displayed “CI/CD Pipeline completed successfully!” and “Finished: SUCCESS”.
- Two successful builds were recorded in Jenkins, confirming repeatable pipeline execution.

7. Conclusion

The basic CI/CD workflow was successfully implemented using GitHub and Jenkins. The workflow demonstrates source-code integration, automated checkout, and sequential Build, Test, and Deploy stages. The successful Jenkins build confirms that the pipeline was executed correctly.